Software Engineering

UE24CS341A

5th Semester, Academic Year 2026

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Lab 1

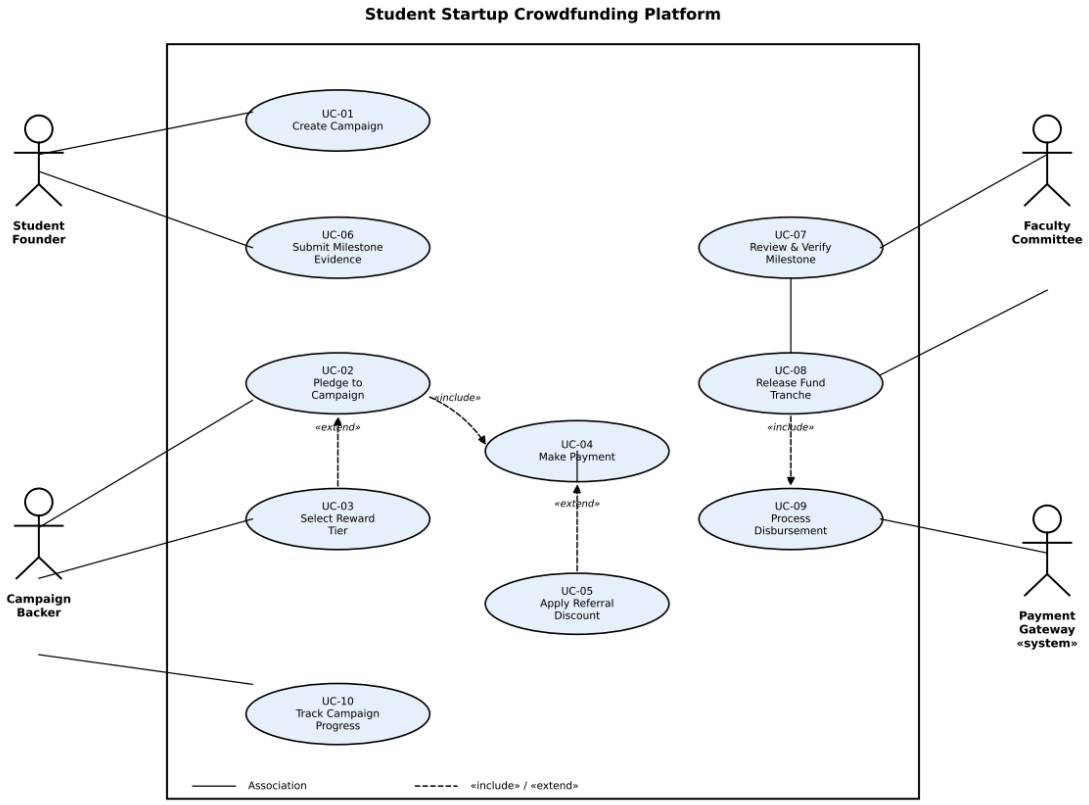
Requirements Table

Actors: Student Founder, Campaign Backer, Faculty Committee, Payment Gateway (external system)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall collect backer pledges and disburse funds to the student venture in tranches only upon verification of predetermined project milestones.	High	Pass: Milestone evidence reviewed and tranche released. Fail: Campaign funds disbursed without milestone sign-off.	Ensures funds are released based on actual project progress.
FR-002	Functional	The system shall allow a Student Founder to create a campaign by specifying a funding target, campaign description, and a milestone schedule of at least two milestones before the campaign can be published.	High	Pass: Publish is blocked and an error is shown if funding target, description, or fewer than two milestones are missing. Fail: Campaign publishes with any field absent.	Campaign creation is the core functionality for student ventures.
FR-003	Functional	The system shall allow a Campaign Backer to select a reward tier when pledging, displaying the reward description and minimum pledge amount for each available tier.	Medium	Pass: Pledge is accepted only when the entered amount meets or exceeds the selected tier's minimum. Fail: Pledge below the tier minimum is accepted.	Backers need campaign information before deciding whether to pledge.

FR-004	Functional	The system shall allow a Student Founder to submit milestone-completion evidence (documents, photos, or progress reports) linked to a specific milestone for Faculty Committee review.	High	Pass: Submitted evidence is time-stamped, linked to the correct milestone ID, and appears in the Faculty Committee's review queue within 1 minute. Fail: Evidence is accepted without a milestone link.	Enables backers to financially support student ventures.
FR-005	Functional	The system shall notify all backers of a campaign, via email and in-app notification, when a milestone is verified and the corresponding fund tranche is released.	Medium	Pass: 100% of a campaign's backers receive a notification within 5 minutes of tranche release. Fail: Notification is missing or sent to backers of a different campaign.	Faculty verification is required before tranche release.
NFR-001	Nonfunctional	The payment processing gateway must comply with PCI-DSS guidelines, using tokenized payment authorizations for every transaction.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load.	Protects sensitive backer payment information.
NFR-002	Nonfunctional	The system shall encrypt all personally identifiable information (PII) and financial records at rest using AES-256, and shall maintain at least 99.5% platform uptime measured monthly.	High	Pass: A security audit confirms AES-256 encryption at rest for all PII/financial fields, and monthly uptime logs show ≥99.5% availability. Fail: Any unencrypted PII field is found, or uptime falls below 99.5%.	Prevents unauthorized fund releases.

UML Use-Case Diagram



Use-Case Flow Specification

Use Case ID / Name	UC-02: Pledge to Campaign
Primary Actor	Campaign Backer
Secondary Actors	Payment Gateway (external system)
Related Use Cases	«include» UC-04 Make Payment | «extend» UC-03 Select Reward Tier
Preconditions	1) The Campaign Backer is logged in. 2) The target campaign is published and currently active (before its funding deadline). 3) The Payment Gateway service is reachable.
Postconditions	Success: The pledge is recorded against the campaign, the campaign's raised-amount total is updated, and the backer receives a pledge confirmation. Failure: No pledge record is created and the campaign total is unchanged.

Main Success Scenario

1. Campaign Backer opens a campaign page and selects “Back this Project.” The system displays the campaign's funding target, amount raised so far, and the list of reward tiers.

2. Campaign Backer selects a reward tier (UC-03 — extends this use case when tiers are offered) or chooses to pledge without a tier.

3. The system displays a pledge form and the Backer enters a pledge amount.

4. The system validates that the amount is at least the campaign's minimum pledge and, if a tier was selected, at least that tier's minimum.

5. The system invokes UC-04 Make Payment («include») to collect and authorize the payment.

6. Campaign Backer selects a payment method and confirms the transaction.

7. The system sends the authorization request to the Payment Gateway.

8. Payment Gateway authorizes the transaction and returns a confirmation token.

9. System records the pledge, links it to the reward tier (if any), and updates the campaign's total amount raised.

10. The system displays a pledge confirmation and emails a receipt to the Campaign Backer.

11. Use case ends successfully.

Alternate Flow: 8a. Payment Declined

- 8a1. Payment Gateway declines the transaction; System displays a “payment failed” message and preserves the entered pledge amount and tier selection.
- 8a2. Campaign Backer is prompted to retry with the same or a different payment method.
- 8a3. If payment fails again after 2 attempts, the pledge is cancelled, no funds are recorded, and the Backer is notified with a suggestion to contact support.